

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

4 Computation Theory (ad260)

Let P be any register machine program.

- (a) What is the partial function $f: \mathbb{N} \rightarrow \mathbb{N}$ of one argument computed by P ? [2 marks]

Register machines

Answer: f is defined by $f(x) \triangleq y$ if y is the value in register R_0 when the program P halts, having been started with the value 0 in register R_0 , x in register R_1 and 0 in all other registers. If P does not halt from this starting configuration, then $f(x)$ is undefined.

- (b) What is the partial function $g: \mathbb{N}^2 \rightarrow \mathbb{N}$ of two arguments computed by P ? [2 marks]

Register machines

Answer: g is defined by $g(x, y) \triangleq z$ if z is the value in register R_0 when the program P halts, having been started with the value 0 in register R_0 , x in register R_1 , y in register R_2 and 0 in all other registers. If P does not halt from this starting configuration, then $g(x, y)$ is undefined.

- (c) Describe the construction of a *Gödel numbering* of register machine programs. That is, a bijection G between the natural numbers $\mathbb{N}$ and the collection of register machine programs. [5 marks]

Universal register machine

Answer: There is more than one way to do this. One Gödel numbering is covered in detail in lectures. We define a coding of pairs of natural numbers $\langle\langle x, y \rangle\rangle \triangleq 2^x(2y + 1)$ and $\langle x, y \rangle = 2^x(2y + 1) - 1$. This enables us to define a coding of lists, by $\ulcorner \square \urcorner \triangleq 0$ and $\ulcorner x :: l \urcorner \triangleq \langle\langle x, \ulcorner l \urcorner \rangle\rangle$.

We then define a function *body* that assigns to each $n \in \mathbb{N}$ a unique register machine instruction by $\text{body}(0) = \text{HALT}$, $\text{body}(\langle\langle 2i, z \rangle\rangle) = R_i^+ \rightarrow L_z$ and $\text{body}(\langle\langle 2i + 1, z \rangle\rangle) = R_i^- \rightarrow L_j, L_k$ where $z = \langle j, k \rangle$.

$$\text{Finally } G(i) = \begin{bmatrix} L_0 & : & \text{body}(x_0) \\ & \vdots & \\ L_n & : & \text{body}(x_n) \end{bmatrix} \text{ where } i = \ulcorner [x_0, \dots, x_n] \urcorner$$

We now write $\phi_i: \mathbb{N} \rightarrow \mathbb{N}$ for the partial function of one argument that is computed by the register machine program $G(i)$, and $\psi_i: \mathbb{N}^2 \rightarrow \mathbb{N}$ for the partial function of two arguments that is computed by the register machine program $G(i)$, where G is the Gödel numbering constructed in part (c).

- (d) Show that the partial function $u: \mathbb{N}^2 \rightarrow \mathbb{N}$ defined by $u(i, x) = \phi_i(x)$ for all i and x is computable. You may assume standard results about register machine programs, as long as you state them in full and clearly. [5 marks]

Universal register machine

Answer: Section 4.3 of the lecture notes describes the construction of a universal machine which, when started with a value 0 in register R_0 , the value i in register R_1 and the value

$\lceil x_1, \dots, x_n \rceil$ in register R_2 with 0 in all other registers, computes the value $f_i^n(x_1, \dots, x_n)$ where f_i^n is the function of n arguments computed by the program $G(i)$.

Thus u is computed by a program which replaces the contents x of R_2 with $\lceil x \rceil$ and then invokes the machine U .

Alternatively, the construction of a machine computing u can be described explicitly along the lines of the construction of U .

- (e) Sketch a proof to show that there is a computable partial function $s: \mathbb{N}^2 \rightarrow \mathbb{N}$ such that, for all $x, y, z, \in \mathbb{N}$:

$$\phi_{s(x,y)}(z) = \psi_x(y, z).$$

[6 marks]

Universal register
machine,
Undecidability

Answer: $s(x, y)$ can be thought of as a function that takes the code (x) of a program P and a value y and produces the code of a program P_y which takes a single argument z , copies it into the second register, inserts y in the first register and then invokes P .

We can describe a register machine that behaves exactly as such. Some details of such a description will be expected in the answer.
